



PENIEL N. ROCHA

INFORMATION TECHNOLOGY

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🌐 <https://portfolio-jcfsysdev.vercel.app>

About me

Compassionate and highly driven individual with experience in technical support, system troubleshooting, software deployment, and IT operations. Skilled in Python, Java, JavaScript, web development, database management, and machine learning integration, with hands-on experience gained through academic and technical projects.

System Development

GRAINSCAN: Automated Rice Quality Inspection with Image Processing and Data Analytics

GrainScan is an automated system for rice quality inspection that uses image processing and data analytics to evaluate grains based on length, discoloration, and breakage.

- **Python** – Core programming language used for system development and model integration.
- **YOLOv8** – Object detection and segmentation of rice grains in captured images.
- **OpenCV** – Image processing, feature extraction, and grain analysis.
- **TensorFlow/PyTorch** – Building, training, and deploying machine learning models for rice inspection.

MoodSync: AI

MoodSync AI is an intelligent music recommendation system that personalizes Spotify playlists based on users' detected moods.

- **Python** – Core programming language used for emotion detection, recommendation logic, and system integration.
- **Flask** – Web framework for developing the application and handling user requests.
- **Spotify API** – Accessing music data, playlists, and song recommendations.
- **OpenCV** – Capturing and processing facial images for emotion detection.
- **TensorFlow** – Building, training, and deploying emotion recognition models.
- **HTML/CSS/JavaScript** – Frontend development and user interface design.

Barcode Integration in Library System Enhancing Accessibility and Efficiency in Resource Management

This system enhances library accessibility and efficiency through the integration of barcode technology into a library management system.

- **HTML/CSS/JavaScript** – Frontend development and user interface design.
- **PHP** – Backend development and business logic implementation.
- **MySQL** – Database management for student records.

Internship Experience

Technical Support Intern

Ttec| January 2026 – May 2026

- Provided technical assistance and troubleshooting for hardware, software, and network-related issues.
- Assisted users in diagnosing and resolving operating system and application problems.
- Performed system reimaging and software installation on Windows devices.
- Configured and updated workstations to ensure optimal performance.
- Supported basic network connectivity troubleshooting and device setup.
- Maintained accurate documentation of technical issues and resolutions.
- Collaborated with IT personnel to ensure timely issue resolution and service delivery.

Certifications

- Cisco Networking Academy - CCNA: Switching, Routing, and Wireless Essentials
- Cisco Networking Academy - Cyber Threat Management
- Cisco Networking Academy - Networking Devices and Initial Configuration
- Cisco Networking Academy - CCNA: Introduction to Networks
- DataSense Analytics - Data Analytics Fundamentals
- Information Technology Specialist - Python

Details

Date of Birth: Jan. 22, 2002
Languages: English, Filipino
Nationality: Filipino

Education

Bachelor of Science in Information Technology

Our Lady of Fatima University
2022-2026

Senior Highschool

Our Lady of Fatima University
2019-2021

Skills

- **Programming Languages**
 - Python
 - C/C++
 - Java
 - JavaScript
- **Web Development**
 - HTML
 - CSS
 - PHP
 - Bootstrap
 - React
- **Networking**
 - Basic Network Troubleshooting
 - IP Configuration
 - Routing and Switching
- **System & AI Development**
 - Machine Learning Integration
 - System Testing
 - System Deployment
 - Image Annotation/Data Labeling
 - Train/Fine-tune Model
- **IT Support & Operations**
 - System Diagnostics
 - Hardware/Software Troubleshooting
 - Operating System (Windows/macOS)
 - Hardware Maintenance & Upgrades
- **Other Skills**
 - Photography
 - Videography
 - VideoEditor
 - Audio Mixing
 - Musician
 - Graphic Design